Worksheet 7
Actuarial Corporate Finance

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1

We are given a Sharpe ratio of 0.4 for the market, as well as a Sharpe ratio of 0.33 for the QRS stock, a correlation of 0.8 of the stock with the market, and a volatility of 0.3. We want to find the alpha for the stock. The equation for α is given by

$$\alpha_{QRS} = \mathbb{E}\left[R_{QRS}\right] - r_{QRS}$$

where the return r_{QRS} is given by

$$r_{QRS} = r_f + \beta\left(\mathbb{E}\left[R_M\right] - r_f\right)$$

The Sharpe ratio is given by $\frac{\mathbb{E}[R_{QRS}]-r_f}{\sigma_{QRS}}$ so we can write

$$0.33 = \frac{\mathbb{E}\left[R_{QRS}\right] - r_f}{\sigma_{QRS}}$$

We can find the expected return by using the correlation to find the β for the stock, which will then yield the expected return via the CAPM. We have $\beta_{QRS} = \rho\frac{\sigma_{QRS}}{\sigma_M} = 0.8 * \frac{0.3}{\sigma_M}$, but this still needs a volatility for the market. But by the market's Sharpe ratio we know

$$0.4 = \frac{\mathbb{E}\left[R_M\right] - r_f}{\sigma_M}$$

Substituting everything, we can write

$$\alpha_{QRS} = \mathbb{E}\left[R_{QRS}\right] - r_f - \beta\left(\mathbb{E}\left[R_M\right] - r_f\right) = \mathbb{E}\left[R_{QRS}\right] - r_f - \rho\frac{\sigma_{QRS}}{\sigma_M}(\mathbb{E}\left[R_M\right] - r_f)$$

Looking at this, we can see that the right-hand quantity contains the Sharpe ratio of the market! The first term of the right-hand side can also be transformed into a Sharpe ratio if we divide through by σ_{QRS}:

$$\frac{\alpha}{\sigma_{QRS}} = \frac{\mathbb{E}\left[R_{QRS}\right] - r_f}{\sigma_{QRS}} - \rho SR_M = SR_{QRS} - \rho SR_M$$

Now we can substitute numbers:

$$\frac{\alpha}{0.3} = 0.33 - 0.8 \times 0.4 \implies \boxed{\alpha = 0.003}$$

2

We are given a stock price of 60 and future earnings of 10%. No other fundamentals have changed. What happens to the stock price? We can set the stock price as

$$S_0 = PV(Div_t)$$

where the right-hand side is the present value of future dividends. Assuming the dividend is a perpetuity (i.e. dividend payments happen forever), dividends across time are constant, and no probability of the company failing, we have $PV(\sum(Div_t)) = \frac{Div}{r}$, which gives

$$S_1 = \frac{Div}{r} = S_0$$

If we then assume dividends go up by 1.1 in conjunction with the future earnings increasing, we then have

$$S_1 = \frac{1.1Div}{r} = 1.1S_0$$

so the stock price goes up 10% to $\boxed{S_1 = 66}$.

3

Suppose ABC and XYZ are both expected to pay 2.2 million dividends into perpetuity (per year). ABC's cost of capital is 12%; XYZ's is 16%. We recall that we could set the return equal to the cost of capital.

3.1

Which has the higher market value? We write $\frac{D}{r_i} = V_i$ so it immediately follows that since $\frac{D}{r_1} = \frac{D_{ABC}}{0.12}$ is higher, it must have higher market value V_1.

3.2

As said before, firm XYZ should have higher expected return since it is equal to the cost of capital.

3.3

By definition we know

$$\alpha_s = \mathbb{E}[R_s] - r_s$$

which means that XYZ has the higher α since $r_{XYZ} = 16\% - 15\% = 1\%$ while $\alpha_{ABC} = -3\%$.

3.4

The logic of the previous problems immediately implies that a higher α implies a lower market value, since market value is inversely related to expected return.